- TC & SC ✗
- Asymptotic Analysis ✗
- Big O ✗
- TLE ✗

\# of ops
\# of its

Quiz 1 :

Sum of first N natural no's.

$1 + 2 + 3 + \cdots + N = \frac{N(N+1)}{2}$

Quiz 2 :

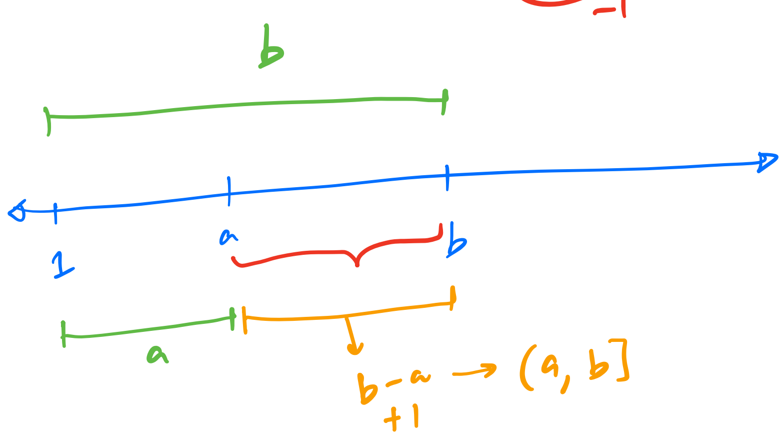
$[3, 10] \rightarrow 3, 4, 5, 6, 7, 8, 9, 10 \rightarrow 8$

[ ] → Inclusive

( ) → Exclusive

[a, b]	[a, b)	(a, b)
b − a + 1	b − a	b − a − 1

−1 −1



Q void fun (int N) {

```
    S = 0
    f (i = 1; i <= N; i++) {
        S = S + i;
    }
    return S;
}
```

$i = 1, 2, 3, \ldots, N$

$i : [1, N]$

$\boxed{\#it \to N}$

Q

```
f (i = 0; i <= 100; i++) {
    print(10);
}
```

#it

$i = 0, 1, 2 \ldots 100$

$i : [0, 100]$

$\boxed{\#it : 101}$

Q

```
void func(int N, int M) {
    f(i=1; i<=N; i++) {
        if(i%2==0) {
            print(i);
        }
    }

    f(i=1; i<=M; i++) {
        if(i%2==0) {
            print(i);
        }
    }
}
```

i: [1, N]

\# it → N

i: [1, M]

\# it → M

$\boxed{\#it \rightarrow N+M}$

Q

```
int func(int N) {
    s=0
    f(i=1; i<=N; i=i+2) {
        s=s+i;
    }
}
```

i=1, 3, 5, 7, 9 -- N

\#it → ~ N/2

Q

```
void func(N) {
    f(i=1; i*i <= N; i++) {
        S = S + i^2
    }
    return S;
}
```

$i = 1, 2, 3, \ldots, \boxed{\sqrt{N}}$

$[1, \sqrt{N}]$

$\boxed{\#it \to \sqrt{N}}$

$i \times i <= N$

$\sqrt{i^2} <= \sqrt{N}$

$\boxed{i <= \sqrt{N}}$

Basics of Log

$\log_b a \to$ Log a base b : to what power do we raise b to get a.

$$\boxed{\log_b a = c \Rightarrow b^c = a}$$

- $\boxed{\log_2(16) \to 4}$ $\quad 2^{\underline{4}} = 16$
- $\boxed{\log_2(64) \to 6}$ $\quad 2^{\underline{6}} = 64$
- $\boxed{\log_3(81) \to 4}$ $\quad 3^{\underline{4}} = 81$
- $\boxed{\log_2(2^{10}) \to 10}$ $\quad 2^{\underline{10}} = 2^{10}$

27 /2

13 /2

6 /2

3 /2

1

$$\boxed{\log_2 2^K = K}$$

Q

```
void func(N) {
    i = N;
    while (i > 1) {
        i = i/2;
    }
}
```

STEP

STEP		
	$i : N$	
1	$i : N/2$	$: N/2^1$
2	$i : N/4$	$: N/2^2$
3	$i : N/8$	$: N/2^3$
K	$\boxed{i : 1}$	$: N/2^K$

$$N/2^K = 1$$

$$2^K = N$$

take log on both sides

$$\log_2(2^K) = \log_2(N)$$

$$\boxed{K = \log_2(N)}$$

N vs	$\sqrt{N}$ vs	$\log_2 N$
10^6	10^3	~ 20

$2^{10} = 1024 \sim 10^3$

$2^{20} \longleftrightarrow 10^6$

$2^{30} \longleftrightarrow 10^9$

$$2^{20} \cdot 2^4$$

$$\boxed{10^6 \quad 16}$$

$\log_2(N)$

$$N \xrightarrow{/2} N/2 \xrightarrow{/2} N/4 \xrightarrow{/2} \cdots \longrightarrow 1$$

$$1 \xrightarrow{\times 2} 2 \xrightarrow{\times 2} \cdots N/4 \xrightarrow{\times 2} N/2 \xrightarrow{\times 2} N$$

Q

```
void func(N) {
    s=0
    f(i=1; i<=N; i=i*2){
        s=s+i;
    }
}
```

it	i	
1	1	×2
2	2	×2
3	4	×2
4	8	×2
5	16	
⋮	⋮	×2
	$>N$	

$$\boxed{\# it \rightarrow \log_2(N)}$$

Q

```
void func(N)
    s=0
    f(i=0; i<N; i=i*2){
        s=s+i;
    }
}
```

#it	i
1	0
2	0
3	0
4	0
⋮	⋮
∞	0

```
void func(N) {
    f(i=1; i<=10; i++){
        f(j=1; j<=N; j++){
            print(..);
        }
    }
}
```

i	j : [1, N]	# it
1	[1, N]	N
2	[1, N]	+ N
3	[1, N]	+ N
⋮	⋮	+ ⋮
10	[1, N]	+ N

$\boxed{\# it \to 10 . N}$

```
void func(N) {
    f(i=1; i<=N; i++)
        f(j=1; j<=N; j++){
            print(i*j);
        }
    }
}
```

i	j : [1, N]	# it
1	[1, N]	N
2	[1, N]	+ N
⋮	⋮	+ ⋮
N	[1, N]	+ N

$\boxed{\# it \to N^2}$

Q

```
void func(N) {
    for (i=1; i<=N; i++) {
        for (j=1; j<=N; j=j*2) {
            print(i*j);
        }
    }
}
```

i	$j : [1, N]$	# it
1	1, 2, 4, ... N	$\log_2 N$
2	1, 2, 4, ... N	+ $\log_2 N$
⋮	⋮	+ ⋮
N	1, 2 — N	+ $\log_2 N$

$\boxed{\#it \rightarrow N \log N}$

AP: Arithmetic Progression

Series : 4, 7, 10, 13, 16, 19, 22, 25

$\underbrace{4, 7}_{3}, \underbrace{7, 10}_{3}, \underbrace{10, 13}_{3}, \underbrace{13, 16}_{3}$ - ——— ———

General : a, $a+d$, $a+2d$, $a+3d$, $a+4d$, ———, $a+(n-1)d$

1 2 3 4 5 ... N

$$\boxed{\text{Sum of } AP_N = \frac{N}{2}\left[2a+(n-1)d\right]}$$

a : first term

d : common diff

n : no. of terms.

4, 7, 10, ———

$\frac{10}{2}\left[2\times 4+(10-1).3\right]$

(4) GP: Geometric Progression

Eg: 3, 6, 12, 24, 48 - ———

$3 \xrightarrow{\times 2} 6 \xrightarrow{\times 2} 12 \xrightarrow{\times 2} 24$ — ——— ———

General : a $a\times r$ $a\times r^2$ $a\times r^3$ ——— - $a.r^{n-1}$

1 2 3 4 - - N

$$\boxed{\text{Sum of first } N \text{ terms of GP} : a\left[\frac{r^n-1}{r-1}\right]}$$

$\boxed{r \neq 1}$

AP : d=0

→ 3 3 3 3 ———

$r=1$

$\boxed{r==1}$

$\boxed{a.n}$

Q

```
void func(N) {
    f(i=1; i<=2^N; i++) {
        print(i);
    }
}
```

Q

```
void func(N) {
    f(i=1; i<=N; i++) {
        f(j=1; j<=2^i; j++) {
            print(i*j);
        }
    }
}
```

i	j : [1, 2^i]	#it
1	[1, 2^1]	2^1
2	[1, 2^2]	+ 2^2
3	[1, 2^3]	+ 2^3
|		+ |
N	[1, 2^N]	2^N

$$\underbrace{2^1 + 2^2 + \text{———} + 2^N}_{} \quad : \text{GP}$$

$a = 2 \quad r = 2 \quad n = N$

$$a\left[\frac{r^n - 1}{r-1}\right] = 2\left[\frac{2^N - 1}{2-1}\right] = 2[2^N - 1]$$

```
f( i=1; i<=4; i++){
    f(j=1; j<=i; j++){
        ~~~
    }
}
```

i	j:[1, i]	#it
1	[1, 1]	1
2	[1, 2]	+ 2
3	[1, 3]	+ 3
4	[1, 4]	+ 4

$\boxed{\#it \rightarrow 10}$

$\frac{N}{2}[N+1]$

$\frac{n}{2}[2a+(n-1)d]$ $\quad \frac{N}{2}[2\times1+(N-1)\times1]$

```
void func(N){
    f(i=0; i<N; i++){
        f(j=0; j<=i; j++){
            print(i*j);
        }
    }
}
```

i	j:[0, i]	#it
0	[0, 0]	1
1	[0, 1]	+ 2
2	[0, 2]	+ 3
|		+ |
|		|
N-1	[0, N-1]	N

AP: $a=1$
$d=1$
$n=N$

$1+2+3+\text{—}\ N$

Q

```
f(i=N; i>0; i=i/2){
    f(j=1; j<=i; j++){
        print(i*j);
    }
}
```

i	j : [1, i]	# it
N	[1, N]	N
N/2	[1, N/2]	+ N/2
N/4	[1, N/4]	+ N/4 +
|		|
1	[1, 1]	1

K

$$\#it \rightarrow \underbrace{N + N/2 + N/4 + \cdots + 1}_{\times 1/2}$$

I

$a : N$

$r = 1/2$

$n : \log_2 N$

$$\frac{a[r^n - 1]}{r - 1}$$

II

$$N + N/2 + N/4 + \cdots + 1$$

$$\frac{N}{2^0} + \frac{N}{2^1} + \frac{N}{2^2} + \cdots + \frac{N}{2^k}$$

$$\frac{N}{2^k} = 1 \rightarrow \boxed{N = 2^k}$$

$$\frac{2^k}{2^0} + \frac{2^k}{2^1} + \frac{2^k}{2^2} + \cdots + \frac{2^k}{2^k}$$

$$\boxed{2^k + 2^{k-1} + 2^{k-2} + \cdots 2^2, 2^1, 2^0}$$

$a = 2^0$

$n = K$

$r = 2$

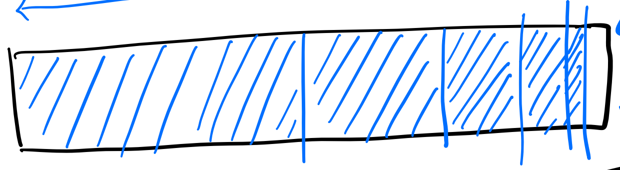
$$= 2^{k+1} - 1 = 2^k \cdot 2 - 1 = 2N - 1$$

III

$$N \mid \underbrace{+ \frac{N}{2} + \frac{N}{4} + \cdot + \cancel{\ } + \cancel{\ } + 1}$$

$N +$

N



$N-1$

$2N-1$

$N = 16$

16

$$\frac{16}{2} + \frac{16}{4} + \frac{16}{8} + \frac{16}{16}$$

Compare

$$\log_2 N < \sqrt{N} < N < N\log N < N\sqrt{N} < N^2 < N^3 < 2^N$$

20 10^3 10^6 2×10^7 10^9 10^{12} 10^{18} $\boxed{2^{10^6}}$

HW

$N!$

$N \times (N-1) \times - - 1$

$\boxed{2^{10^6}}$

$\boxed{10^{-}}$

2^{10}

10^3

$N = 10^6$

$$\boxed{2^{10^6} \sim 10^{3\times10^5}}$$

2^{10}

$10 \longrightarrow 3$

$10^6 \longrightarrow$ —

④ Asymptotic Analysis

↳ Analyse algorithms for large input values!

→ Big O Notation

STEPS:

1. Calculate the # of iterations / ops.
2. Neglect the lower order terms.
3. Neglect constant coefficient!

$\# ops = 3N^2 + 10N + 6$

$f(N) = 3N^2 + 10N + 6$

Order: 2, 1, 0

$\underbrace{1 \quad 0}_{\text{Lower order terms}}$

$3N^2$

Constant co-efficient

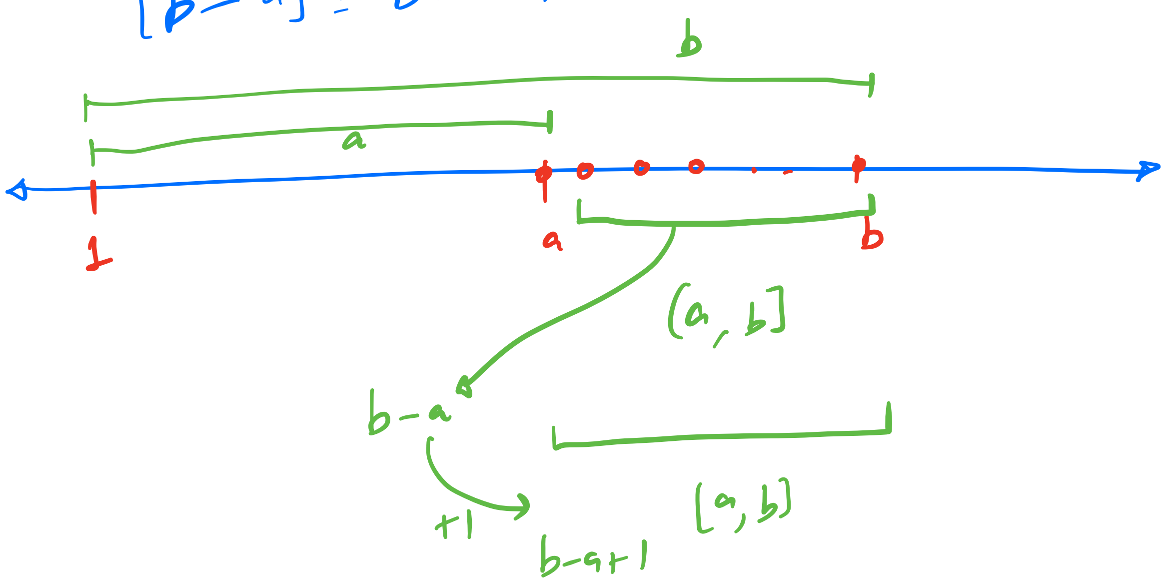
$O(N^2)$

Q $f(N) = 4N^2 + 3N^2\sqrt{N} + 56N\lg N$

$3N^2\sqrt{N}$

$O(N^2\sqrt{N})$

$[b-a] = b-a+1$



$2^{30} \longrightarrow 10^{9}$

$2^{10^6} \longrightarrow 10^{3\times 10^5}$

$2^{10} \longrightarrow 10^{3}$

$2^{20} \longrightarrow 10^{6}$

$2^{30} \longrightarrow 10^{9}$

10 20 40 80 160 —

10 20 40 80

×2 ×2 ×2

✗ AP

✓ GP